

BTEC Level 3 Applied Science • Unit 1 Chemistry • Worksheet**Ionic Bonding**

Original JDScience practice. These questions were written for this resource and are not official exam-board items.

A Retrieval

1. Define ionic bonding.
2. Between which types of element does ionic bonding usually form?
3. Why does solid sodium chloride not conduct electricity?

B Knowledge

4. Describe, in terms of electrons, how NaCl forms.
5. Explain why ionic compounds have high melting points.
6. Why can molten or aqueous ionic compounds conduct electricity?

C Application

7. Deduce formulae for the compounds of Na and O; Mg and Cl; Al and O.
8. Explain why an ionic crystal is brittle.
9. Compare the particles that carry charge in molten NaCl and in a copper wire.

D Exam-style practice

10. Describe how ionic bonding arises in magnesium oxide. [4]
11. Explain why sodium chloride has a high melting point but does not conduct as a solid. [4]
12. Deduce the formula of the compound formed between Al^{3+} and SO_4^{2-} . [2]

E Challenge

13. Suggest why magnesium oxide has a higher melting point than sodium chloride, referring to charge and attraction.